

PII Redaction Tool: Evaluation Report

Khushi Gupta

B.Tech CSE (AIML), 4th year, UPES Dehradun

khushiguptaa04@gmail.com | +91 XXXXX XXXXX

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1. At a glance

Evaluation set	Precision	Recall	F1	Accuracy*
Red Herring Prospectus, held-out test set (75 paragraphs)	98.4%	96.9%	97.7%	98.7%
5 unseen prospectuses, redaction-level (100 paragraphs)	100.0%	88.1%	93.7%	–
5 unseen prospectuses, strict (type must match)	88.9%	76.2%	82.1%	97.5%
Synthetic ticket log (18 tickets, with traps)	96.7%	93.5%	95.1%	97.8%
Injection test (180 planted values)	–	93.9%	–	–

*Token-level accuracy, see Section 4. Leak check on the final output: 0 of 261 replaced values remain; all 8 images replaced.

2. Evaluation methods

No labelled data came with the assignment, so I combined methods that each cover a gap left by the others.

Method	What it does	Why it matters	Result
Hand-labelled prospectus sample	200 paragraphs of the Red Herring Prospectus labelled by me; 125 used while building (dev), 75 never looked at (test)	Real precision and recall on the target document	Test: P 98.4%, R 96.9%
5 unseen prospectuses	Tool run <i>unchanged</i> on 5 new prospectuses; 20 paragraphs each labelled	Shows whether the rules generalise beyond one file	P 100%, R 88.1% (redaction-level)
Synthetic ticket log	18 support tickets I wrote, with SSNs, cards, IPs, DOBs, PAN, Aadhaar, GSTIN and traps	Covers PII types absent from the prospectus and tests the ticket-log use case	P 96.7%, R 93.5%
Injection test	Known fake PII inserted into 30 real prospectus paragraphs	Recall inside real text, with labels known automatically	169/180 caught
Traps	Order and ticket numbers, a non-Luhn 16-digit number, a version string, a mistyped Aadhaar, filing dates	Proves precision: look-alikes must not be redacted	All 5 trap tickets left intact
Leak, image and structure checks	Output re-read for any replaced value; images and paragraph/table counts compared	Guards against leaks and broken formatting	0 leaks; 8/8 images; 1006/76 unchanged

3. How I built the ground truth

- **Stratified sampling with fixed seeds.** From each document I drew PII-dense paragraphs (chosen by a keyword filter independent of the tool), random short paragraphs (table cells, headings) to catch false positives, and long prose where NER mistakes are most likely.
- **One labelling rule-book** for every set (Table 1 of the README): people, contact details, addresses, IDs, private companies and their brands are PII; regulators, legal terms, prices, filing dates and order/ticket

numbers are not.

- **Dev/test split.** I tuned only on the dev set. The test set and the 5 unseen prospectuses were never used to change the code.
- **Synthetic and injected data** have labels by construction, because I wrote or planted every value.

4. Metrics

- **True positive (TP):** a prediction with the same PII type as a labelled entity whose text overlaps it. **FP:** a prediction matching nothing. **FN:** a labelled entity that was missed.
- **Precision** = $TP/(TP+FP)$: how much of what I redacted was really PII. **Recall** = $TP/(TP+FN)$: how much of the real PII I caught. **F1** = their harmonic mean. All are micro-averaged over types.
- **Redaction-level** scores ignore the type: they ask only whether the PII was replaced, which is what matters for privacy.
- **Accuracy** has no natural definition for span detection, so I report *token-level accuracy*: the share of words correctly marked PII / not PII. It is inflated because most words are not PII (redacting nothing would score 88.0% on the test set), so precision and recall are the numbers to read.

5. Results

5.1 Red Herring Prospectus (KSH International)

Test set, by type	TP	FP	FN	Precision	Recall	F1
Person	22	0	0	100.0%	100.0%	100.0%
Email	16	0	0	100.0%	100.0%	100.0%
Company	12	1	0	92.3%	100.0%	96.0%
Phone	5	0	0	100.0%	100.0%	100.0%
URL	5	0	0	100.0%	100.0%	100.0%
Address	3	0	2	100.0%	60.0%	75.0%
Test set, all (held-out)	63	1	2	98.4%	96.9%	97.7%
Dev set, all (used while building)	61	6	1	91.0%	98.4%	94.6%

5.2 Five unseen prospectuses (held-out, tool unchanged)

Prospectus (industry)	TP	FP	FN	Precision	Recall	F1
Atomberg Technologies (appliances)	8	0	1	100.0%	88.9%	94.1%
Torrent Gas (utilities)	13	2	3	86.7%	81.2%	83.9%
Vardaan Biotech (biotech)	5	0	2	100.0%	71.4%	83.3%
NoPaperForms Solutions (edtech)	5	2	2	71.4%	71.4%	71.4%
Iberia Pharmaceuticals (pharma) [†]	1	0	2	100.0%	33.3%	50.0%
All 5, strict (type must match)	32	4	10	88.9%	76.2%	82.1%
All 5, redaction-level (type ignored)				100.0%	88.1%	93.7%

[†]Only 3 labelled entities in its sample. All 4 strict false positives are type mismatches (e.g. the family name “Mehta” replaced as the brand of “Mehta Family Trust”), not wrongly redacted text.

5.3 Synthetic ticket log and injection test

Ticket log (18 tickets)				Injection (30 paragraphs)	
Type	TP	FP	FN	Type	Caught
Email, IP (v4 + v6), card, DOB	12	0	0	Email	30/30
SSN, PAN, Aadhaar, GSTIN, URL	5	0	0	Phone	30/30
Person	5	0	1	IP address	30/30
Phone	3	0	1	PAN	30/30
Company	2	1	0	Credit card	26/30
Address	2	0	0	Person	23/30
All	29	1	2	All	169/180

Ticket log: precision 96.7%, recall 93.5%. The mistyped Aadhaar (invalid checksum), order and ticket numbers, the version string and the regulator’s website were all correctly left alone.

6. How the Anjali Labtech prospectus shaped the rules

Before the benchmark, I ran the tool on one extra prospectus (Anjali Labtech, a PDF converted to Word) and compared the output with a list of every identifier visible in it. Its leaks showed failure modes the Red Herring Prospectus never exposed, and each became a *general* rule rather than a document-specific fix.

Failure found	General rule added	Before	After
Partial names left beside fakes; two promoters merged into one	Split name runs at ordinary words (“AND”); grow partial names over neighbouring name words	20/26 names	26/26 names
Emails and links broken across lines by PDF conversion	Match emails and links with a stray space or line break	2/5	5/5
Foreign company forms (“DMCC”, “GmbH”)	Legal-entity suffixes for 100+ countries (cleanco)	2 missed	0 missed
Auditor’s firm registration number	Any ID written after a label (FRN, registration no.)	missed	removed

Legal terms stayed untouched throughout, and the Red Herring test set did not regress. Because these rules were written after seeing Anjali’s errors, its “after” column is not a held-out result; the 5-prospectus benchmark is.

7. Error analysis

Miss or false positive	Cause
Name or link split across two table cells (“Naveen” / “Goyal”)	PDF-to-Word conversion breaks one cell into several paragraphs; I detect per paragraph
Addresses split across lines or with no PIN code	The address rule needs a PIN code or an “Address:” label
Auditor firm without a legal suffix (“Jain & Chopra”)	The company rule relies on a legal suffix; “Chartered Accountants” is not used as a cue yet
Single-mention names (“Devyani Chhajed”, “Rashi Patil”)	spaCy misses the name and it appears nowhere else to propagate from
Bracketed landline “(033) 4509-6991”, spaced mobile “98200 11223”	Formats not in the phone patterns
“CARE” (rating-agency brand) redacted in “CARE Report”	Deliberate choice: recall over precision for a company’s brand

8. Limitations of this evaluation

- One annotator (me); a second annotator and inter-annotator agreement would make the labels more reliable.
- Small samples: 65 test entities and 42 benchmark entities, so per-type and per-document numbers have wide confidence intervals.
- I wrote both the synthetic ticket log and the patterns, so ticket-log recall is likely optimistic; the injection test partly offsets this with random values in real text.
- Dates and financial figures are kept for usefulness; combined with the industry and offer size they could let someone re-identify the issuer through public SEBI filings.
- The benchmark documents were converted from PDF, so conversion quality affects the scores as much as detection does.